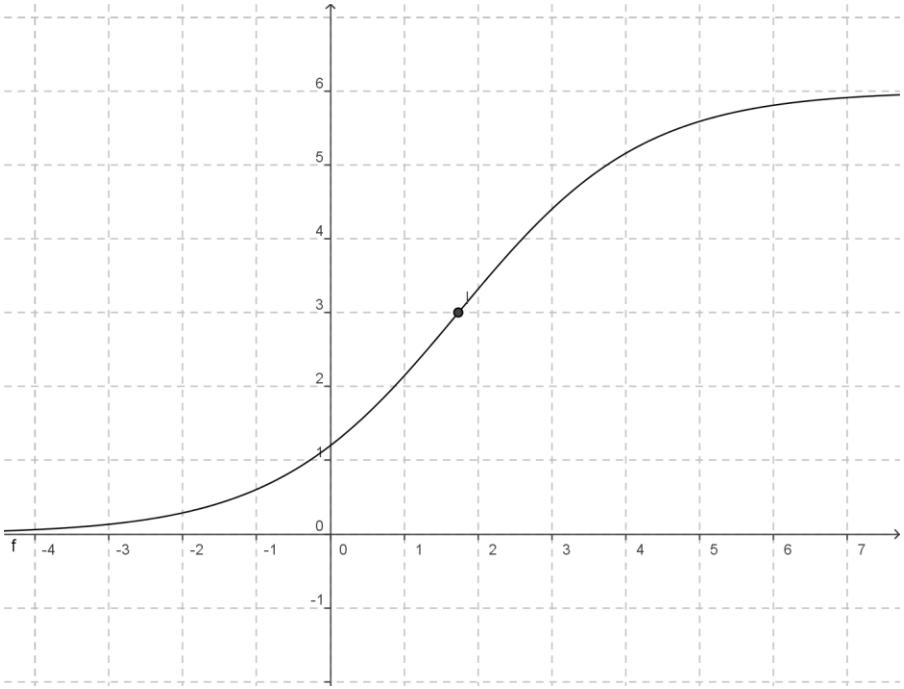


Problem Set 5 Question 3

David Quigley

The Model

- The following logit model is estimated

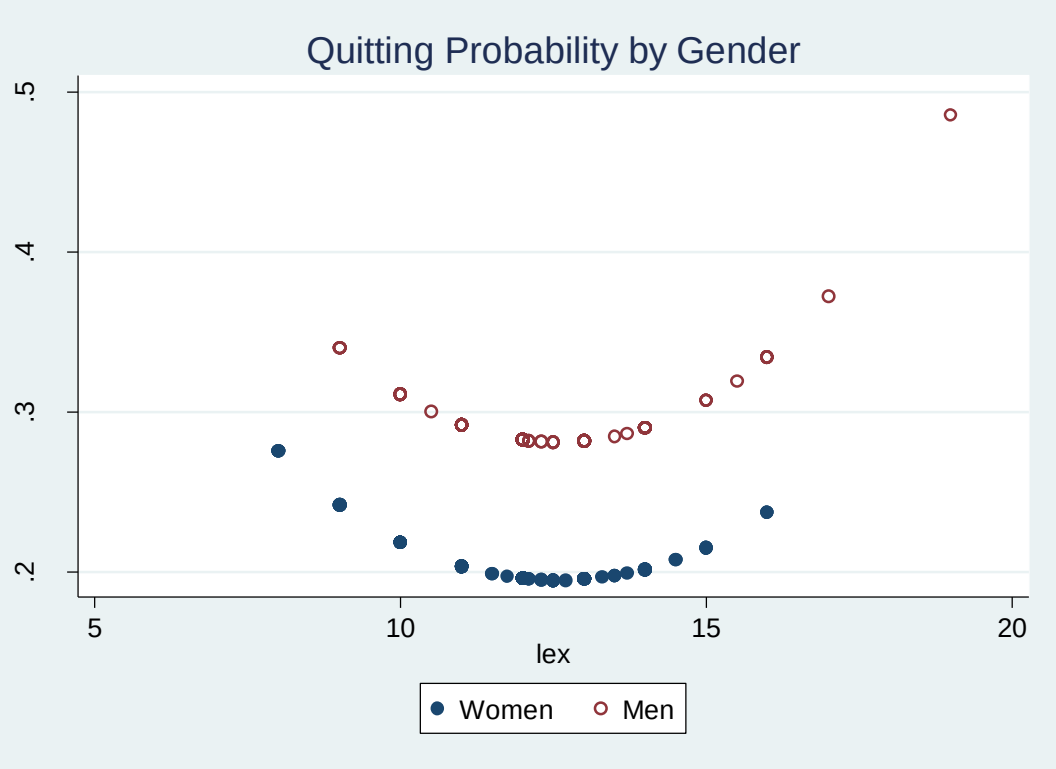
$$logit(P(quit = 1) = (\beta_0 + \beta_1 * sex + \beta_2 * dex + \beta_3 * lex + \beta_4 * lex^2 + u)$$

where *quit* = 1 if the worker quit in the first six months, *dex* is a physical dexterity score, and *lex* is the number of years of education.

Regression Results

Logit regressions				
	(1)	(2)	(3)	(4)
Dep. Var.: Quit(=1)	Logit	Marginal FX	Logit	Logit
Sex(=1)	0.479	0.088	8.268	0.771
	[0.186]**	[0.033]***	[9.394]	[0.697]
Dexterity	-0.094	-0.018	-0.095	-0.164
	[0.014]***	[0.002]***	[0.014]***	[0.133]
Education	-0.539	-0.100	0.768	-0.851
	[0.648]	[0.121]	[1.279]	[1.022]
Education^2	0.021	0.004	-0.039	0.034
	[0.026]	[0.005]	[0.055]	[0.041]
Sex*Education			-1.494	
			[1.555]	
Sex*Education^2			0.07	
			[0.065]	
g^{α}				-0.857
				[2.755]
g^{δ}				-1.838
				[1.999]
Log likelihood	-368.33		-367.15	-361.75
Observations	683		683	683

Regression Results



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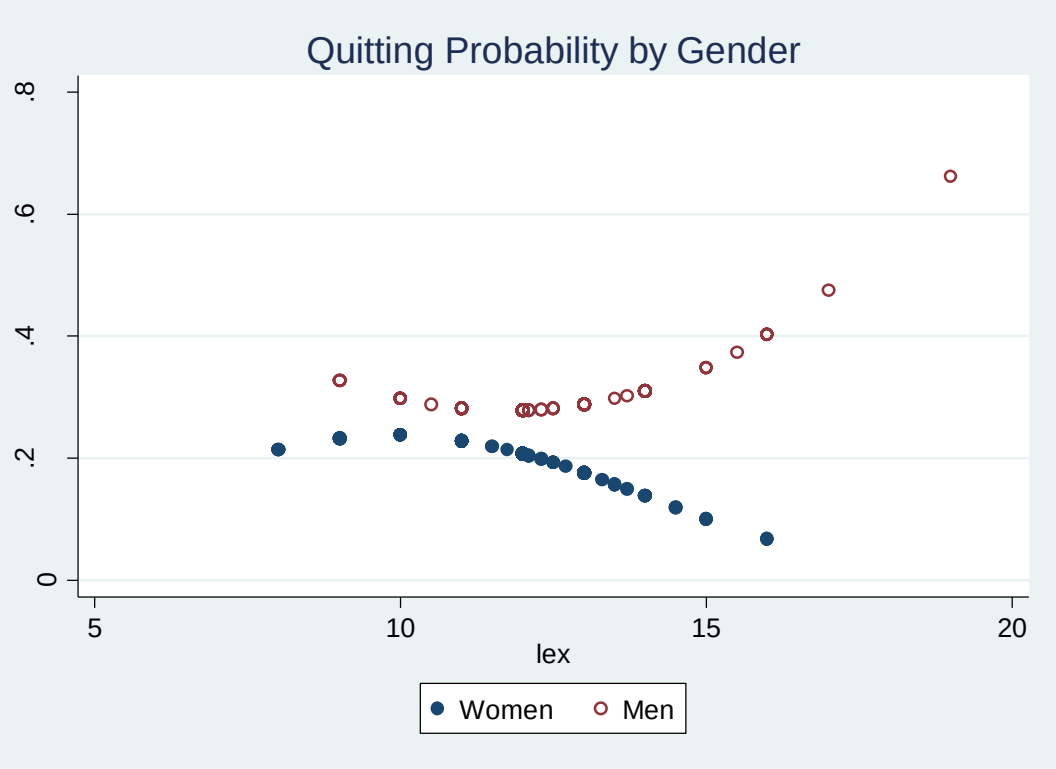

Test Specification

- We test if the logit specification is a good fit using the Pergibon diagnostic
- The Pergibon diagnostic uses a Box-Cox transformation and generates g^{α} and g^{δ}. We add these to the logit regression and test if their coefficients are jointly different from 0.
- Rejecting the null hypothesis ($g^{\alpha} = g^{\delta} = 0$) suggests the logit specification is a bad fit

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Test Specification

- The results of the test are a Chi-squared statistic of 13.01 which has p-value = 0.0015
- The null hypothesis is rejected suggesting the logit specification is a bad fit

Conclusion

- If the firm wants to minimize the probability of workers quitting, the firm should focus on hiring women.
- Assuming the educational effects are the same for men and women, the firm should focus on hiring workers with 11-13 years of education
- However, hiring women with more than 13 years of education should not greatly increase the probability that she quits